

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 28/03/2026**
Time: 3 Hrs **GTM - 07** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

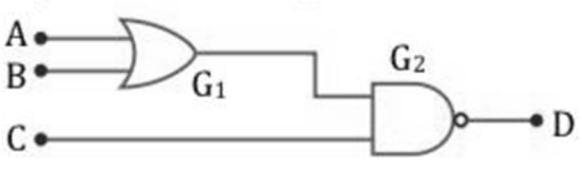
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A particle of mass m carrying charge q is moving in a circle in a magnetic field B. According to Bohr's model, the energy of the particle in the nth level is:

(A) $\frac{1}{n^2}\left(\frac{hqB}{\pi m}\right)$ (B) $n\left(\frac{hqB}{\pi m}\right)$ (C) $n\left(\frac{hqB}{2\pi m}\right)$ (D) $n\left(\frac{hqB}{4\pi m}\right)$

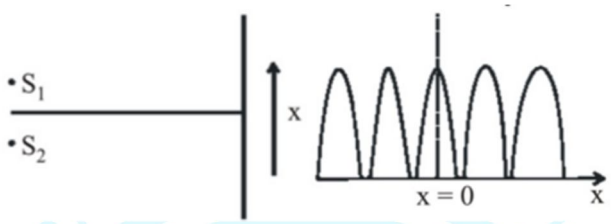
2. For the given combination of gates, if the logic states of inputs A, B, C are as follows $A=B=C=0$ and $A=B=1, C=0$ then the logic states of output D are



A) $0,0$ B) $0,1$ C) $1,0$ D) $1,1$

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3. In a standard YDSE setup, two source S_1 and S_2 of intensity I_1 and I_2 are placed in front of a screen (as shown in figure). The pattern of intensity distribution seen in the central portion is given by graph. In context of the following statements.



(1) S_1 and S_2 must have the same intensities.

(2) S_1 and S_2 have a constant phase difference.

(3) S_1 and S_2 must have the equal phase.

(4) S_1 and S_2 must have the same wavelength.

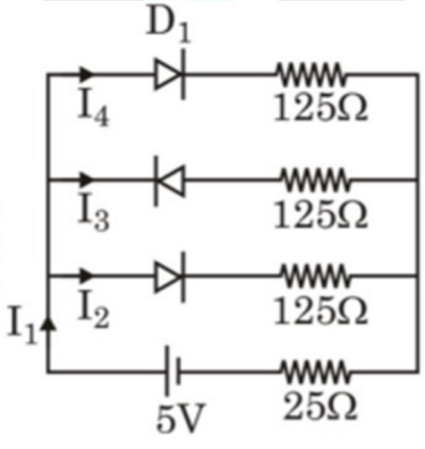
Choose the correct options:

(A) 1 only (B) 4 only (C) 1, 2 & 4 (D) 1 & 2 only

4. At room temperature, the speed of sound in air is 340 m/s. An organ pipe with both ends open has a length L = 29 cm. An extra hole is created at the position L/2. The lowest frequency of sound produced is:

(A) 293 Hz (B) 586 Hz (C) 1172 Hz (D) 2344 Hz

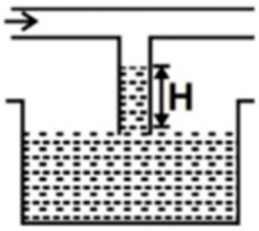
5. If each diode has a forward bias resistance of 75Ω in the below circuit, which of the following options is correct?



(A) $\frac{I_2}{I_3}=1$ (B) $\frac{I_1}{I_2}=2$ (C) $\frac{I_3}{I_4}=1$ (D) $\frac{I_1}{I_2}=1$

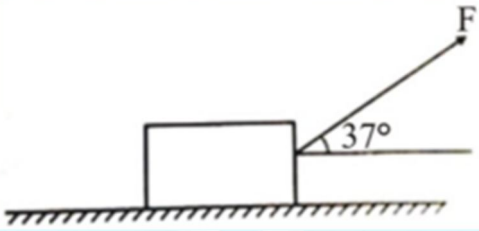
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6. Figure shows a capillary rise H. If air is blown through the horizontal tube in the direction as shown, then rise in capillary tube will become:



(A) = H (B) > H (C) < H (D) zero

7. A 400 N block is dragged along a horizontal surface by an applied force $\vec{F}$ as shown. The coefficient of kinetic friction is μ_k = 0.2 and the block moves with constant velocity. The magnitude of $\vec{F}$ is: (close to)



(A) 100 N (B) 154 N (C) 87 N (D) 290 N

8. When a resistor of 10 Ω is connected across a battery, the current is 0.5A. When a resistor of 5 Ω is connected across the same battery, the current is 0.8A. The emf and internal resistance of the battery are

(A) $10V, \frac{10}{3}\Omega$ (B) $20V, \frac{20}{3}\Omega$

(C) $\frac{20}{3}V, \frac{10}{3}\Omega$ (D) $\frac{40}{3}V, \frac{20}{3}\Omega$

9. A person is swimming with a speed of 10 m/s at an angle of 120° with the flow and reaches to a point directly opposite on the other side of the river. The speed of the flow is '*x*' m/s. The value of '*x*' is:

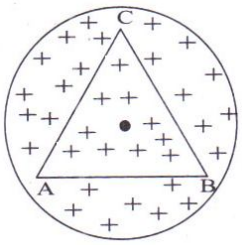
(A) 1 (B) 4 (C) 8 (D) 5

10. A vessel contains an ideal monoatomic gas which expands at constant pressure, when heat of 400 J is given to it. Then the work done in expansion is:

(A) 400 J (B) 240 J (C) 160 J (D) 80 J

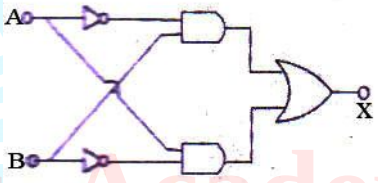
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11. A triangular wire frame having each side equal to 2m is placed in a region of time varying magnetic field having $dB/dt = \sqrt{3} T/s$. The magnetic field is perpendicular to the plane of the triangle as shown in figure. The base of the triangle AB has a resistance 1Ω while the other two sides have resistance 2Ω each. The magnitude of potential difference between the points A and B will be



A) 0.4 V B) 0.6 V C) 1.2 V D) None

12. With reference to the given logic circuit which of the following output is possible for the given inputs A and B



A) A=0, B=0, X=1 B) A=0, B=1, X=0

C) A=1, B=0, X=0 D) A=1, B=1, X=0

13. A resonance tube is resonated with tuning fork of frequency $256Hz$. If the length of first and second resonating air columns are $32cm$ and $100cm$, then end correction will be

A) 1 cm B) 2 cm C) 4 cm D) 6 cm

14. A light beam travelling in +ve x-direction is described by $E_y = 300\sin\left(\omega t - \frac{\omega x}{c}\right) V/m$. An electron is constrained to move along y direction with a speed of $2\times10^7 ms^{-1}$. The maximum magnetic force (in Newton) on the electron is

A) 3.2×10^{-18} B) 5.1×10^{-16} C) 6.5×10^{-11} D) 7.8×10^{-12}

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15. In young's double-slit experiment, the slit separation is 0.5 mm and the screen is 0.5 m away from the slit. For a monochromatic light of wavelength 500 nm, the distance of 3rd maximum from the 2nd minima on the other side of central maxima is

A) 2.75 mm B) 2.5 mm C) 22.5 mm D) 2.25 mm

16. The volume of a liquid (V) that passes through a given cross-section (area A) in a given time (t) depends on velocity of flow (u), as $V = u^a A^b t^c$.Find the correct relation

A) a = c B) a + c = 0 C) 2b + a = 1.5 D) 2b + a + 3 = 0

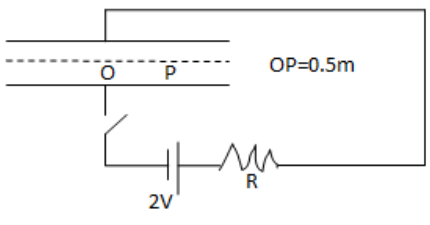
17. A balloon starts rising from the surface of the earth. The ascent rate is constant and equal to v_0. Due to wind the balloon gathers a horizontal velocity component $V_x = Ay$, where A is constant and 'y' is height of ascent, the horizontal drift of the balloon after an ascent 'y' is -

A) $\frac{Ay^2}{2v_0}$ B) $\frac{Ay}{v_0^2}$ C) $\frac{A}{v_0}$ D) $\frac{2Ay^2}{v_0}$

18. A particle of mass m begins to slide down on a fixed smooth sphere from the top. When it breaks off the sphere, it's tangential acceleration will be

A) $2g/3$ B) g C) $\frac{\sqrt{5}}{3}g$ D) $\frac{g}{3}$

19. A parallel plate capacitor with circular plates of radius 1 m has a capacitance of 1 nF. At t = 0, it is connected for charging in series with a resistor R = 1 MΩ across a 2V battery as shown in fig. Calculate the magnetic field at a point P, halfway between the centre and the periphery of the plates, after $t = 10^{-3}$ s.(given $\frac{1}{e} = 0.37$)



A) $8\times10^{-13}T$ B) $8.4\times10^{-14}T$ C) $7\times10^{-14}T$ D) $7.4\times10^{-14}T$

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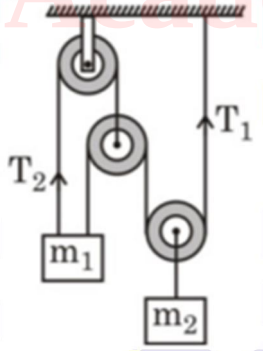
20. An object is projected with a speed 10 m/s at an angle of 30° with the horizontal. The object breaks down into n equal fragments during its motion. One fragment is found to strike the ground at a distance of $\sqrt{3}$ m from the point of projection in the same azimuthal plane, in which the object is projected. If the centre of mass of the remaining fragments strikes the ground simultaneously at distance of $7\sqrt{3}$ m from the point of projection, then the value of n is ($g=10\ m/s^2$)

A) 2 B) 3 C) 4 D) 5

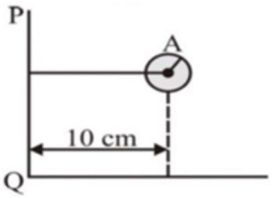
SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. Two blocks of masses m_1 = 5 kg and m_2 = 2 kg are connected by threads which pass over the pulleys as shown in the figure. The threads are massless and the pulleys are massless and smooth. The blocks can move only along the vertical direction. If the acceleration of m_1, if expressed in simplest form, is equal to $\left(\frac{2n}{n-1}\right)$ m/sec^2, calculate 'n'. (take g = 10 m/s^2).



22. Solid sphere A is rotating about an axis PQ. If the radius of the sphere is 5 cm, then its radius of gyration about PQ will be $\sqrt{x}$ cm. The value of x is _____.



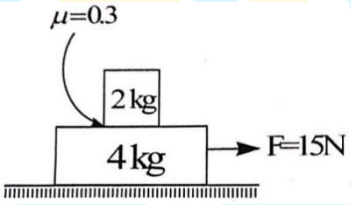
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23. A single slit of width 0.20 mm is illuminated with light of wavelength 500 nm. The observing screen is placed 80 cm from the slit. The width of the central bright fringe (in mm) will be___

24. When a d.c. voltage of 200V is applied to a coil of self inductance $\frac{2\sqrt{3}}{\pi}H$, a current of 1A flows through it. But by replacing d.c. source with a.c source of 200V, the current in the coil is reduced to 0.5A. The frequency of a.c. supply in Hz is

25. Find the frictional force between the blocks in the figure. No friction between 4kg block and ground.

$(g=10ms^{-2})$



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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The variation of equilibrium constant with temperature is given below

Temperature	Equilibrium constant
$T_1 = 25^\circ C$	$K_1 = 10$
$T_2 = 100^\circ C$	$K_2 = 100$

The values of ΔH°, ΔG° at T_1 and ΔG° at T_2 (in kJ mol^{-1}) respectively, are close to

(A) $28.4, -7.14$ and -5.71 (B) $0.64, -7.14$ and -5.71

(C) $28.4, -5.71$ and -14.29 (D) $0.64, -5.71$ and -14.29

27. A commercially sold conc. HCl is 35% HCI by mass. If the density of this commercial acid is 1.46 g/mL, the molarity of this solution is

(Atomic mass : Cl= 35.5 amu, H=1 amu)

(A) 10.2 M (B) 12.5 M (C) 14.0 M (D) 18.2

28. For an elementary chemical reaction, $A_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2A$ the expression for $\frac{d[A]}{dt}$ is

(A) $2k_1[A_2] - k_{-1}[A]^2$ (B) $k_1[A_2] - k_{-1}[A]^2$

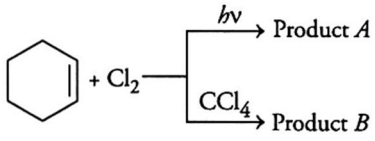
(C) $2k_1[A_2] - 2k_{-1}[A]^2$ (D) $k_1[A_2] + k_{-1}[A]^2$

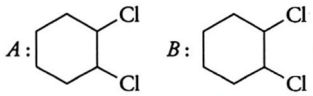
29. Which of the following name formula combination is not correct?

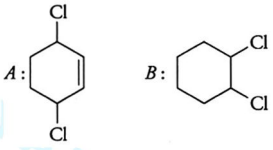
	Formula	Name
(A)	$K_2[Pt(CN)_4]$	Potassium tetracyanoplatinate (II)
(B)	$[Mn(CN)_5]^{2-}$	Pentacyanomanganate ion (II)
(C)	$K(Cr(NH_3)_2Cl_4]$	Potassium diamminetetrachlorochromate (III)
(D)	$[Co(NH_3)_4(H_2O)I]SO_4$	Tetraammineaquaiodido cobalt (III) sulphate

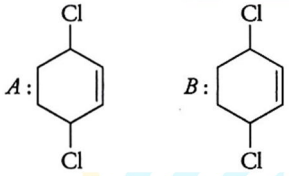
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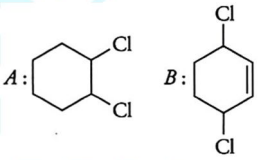
30. Identify product A and product B.



(A) 

(B) 

(C) 

(D) 

31. Consider the following complexes.

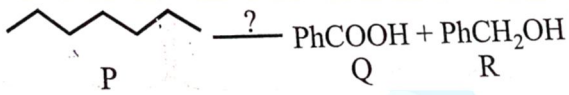
$\underset{(A)}{\left[CoCl(NH_3)_5\right]^{2+}}, \underset{(B)}{\left[Co(CN)_6\right]^{3-}},$

$\underset{(C)}{\left[Co(NH_3)_5(H_2O)\right]^{3+}}, \quad \underset{(D)}{\left[Cu(H_2O)_4\right]^{2+}}$

The correct order of A, B, C and D in terms of wavenumber of light absorbed is

(A) C < D < A < B (B) D < A < C < B

(C) A < C < B < D (D) B < C < A < D

32. 

The correct sequence of reagents for the preparation of Q and R is:

(A) $(i)\,Cr_2O_3, 770K, 20atm$
$(ii)\,CrO_2Cl_2, H_3O^+;$
$(iii)\,NaOH;$
$(iv)\,H_3O^+$

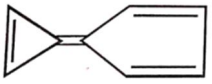
(B) $(i)\,CrO_2Cl_2, H_3O^+$
$(ii)\,Cr_2O_3, 770K, 20\,atm;$
$(iii)\,NaOH; (iv)\,H_3O^+$

(C) $(i)\,KMnO_4, OH^-;$
$(ii)\,Mo_2O_3, \Delta;$
$(iii)\,NaOH;$
$(iv)\,H_3O^+$

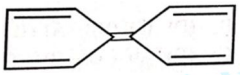
(D) $(i)\,KMnO_4, OH^-$
$(ii)\,CrO_2Cl_2, H_3O^+,$
$(iii)\,NaOH,$
$(iv)\,H_3O^+$

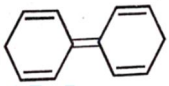
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33. Among the following compounds, the one which shows highest dipole moment is

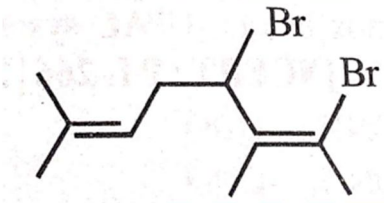
(A) 

(B) 

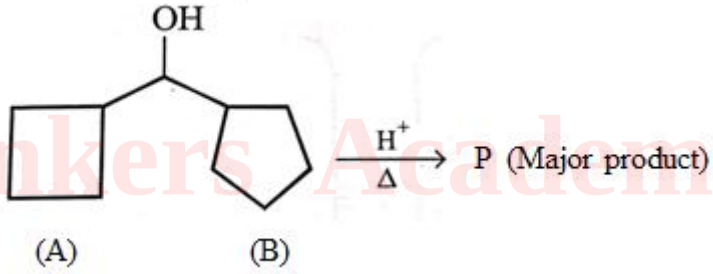
(C) 

(D) 

34. Total number of stereo isomers possible for the given structure:



(A) 8 (B) 2 (C) 4 (D) 3

35. 

In the product P which of the following is observed

(A) Both A and B will convert into six membered rings.

(B) Both A and B will convert into five membered rings.

(C) A converts to six membered and B remains five membered ring

(D) A converts to six membered and B converts to four membered rings

36. Statement-I: The dipole moment of $CH_3 - F$ is greater than that of $CH_3 - Cl$.

Statement-II: Fluorine is more electronegative than Chlorine.

In the light of the above statements, choose the correct answer from the options given below:

A) Both Statement-I and Statement-II are false

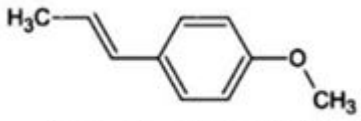
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B) Statement-I is true but Statement-II is false

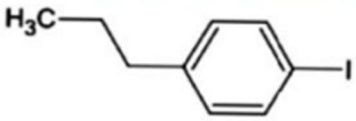
C) Both Statement-I and Statement-II are true

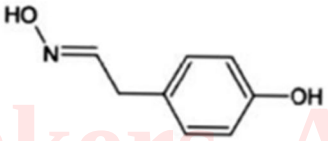
D) Statement-I is false but Statement-II is true

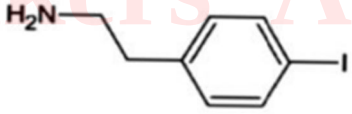
37.

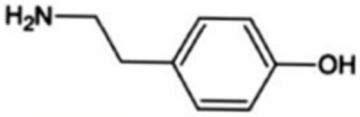


(i) O_3, Zn, CH_3COOH
(ii) CH_3NO_2, OH^- (Δ)
(iii) H_2, Ni
(iv) HI, Δ

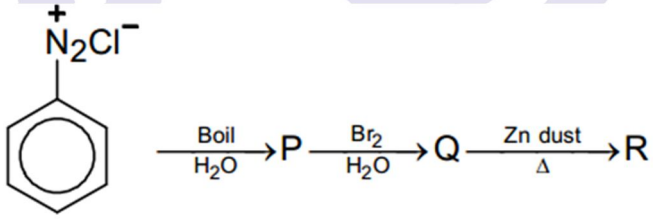
(A) 

(B) 

(C) 

(D) 

38. The dipole moment of final organic product obtained in the following reaction is:



(A) $\mu_R \neq 0$ (B) $\mu_Q = 0$ (C) $\mu_R < \mu_Q$ (D) $\mu_R > \mu_Q$

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39. In the detection of nitrogen, blue/green colour is due to formation of Prussian blue. It is:

(A) $Fe_4^{III}\left[Fe^{II}(CN)_6\right]_3$ (B) $NaFe^{II}\left[Fe^{III}(CN)_6\right]$

(C) $Na_4\left[Fe(CN)_6\right]$ (D) $Na_3\left[Fe(CN)_6\right]$

40. Assertion (A) : The ionic radii of O^{2-} and Mg^{2+} are same.

Reason (R) : Both O^{2-} and Mg^{2+} are isoelectronic species.

In the light of the above statements, choose the correct answer from the options given below.

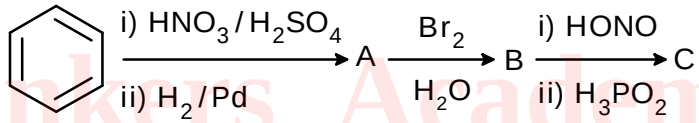
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

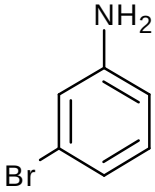
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

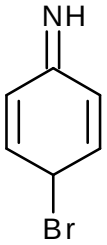
(C) (A) is true but (R) is false

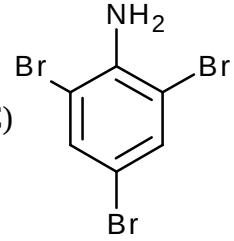
(D) (A) is false but (R) is true

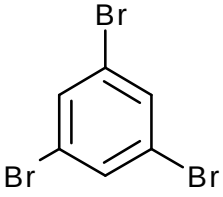
41. The compound C in the reaction sequence is



A) 

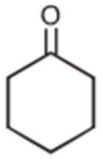
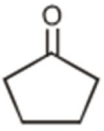
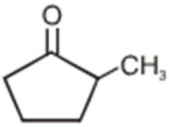
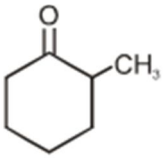
B) 

C) 

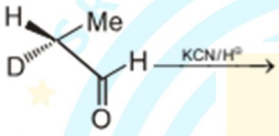
D) 

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42. $HOOC-(CH_2)_5-COOH \xrightarrow{Ca(OH)_2/\Delta} X$, Compound X is

A)  B)  C)  D) 

43. Total number of products formed in the following reaction is



A) 2 B) 3 C) 4 D) 6

44. For H_2S first (Ka_1) and second (Ka_2) ionization constant are 9.1×10^{-8} and 1.2×10^{-13}. What are the concentrations of HS^- and S^{2-} in the mixture $0.1M\,H_2S+0.001M\,HCl$?

A) $9.1\times10^{-8}M, 1.2\times10^{-13}M$ B) $9.1\times10^{-8}M, 1.2\times10^{-11}M$

C) $9.1\times10^{6}M, 1.2\times10^{-18}M$ D) $9.1\times10^{-6}M, 1.092\times10^{-15}M$

45. Pair of transition metal ions having the same spin only magnetic moment is

A) V^{2+}, Co^{2+} B) Ti^{2+}, Co^{2+}

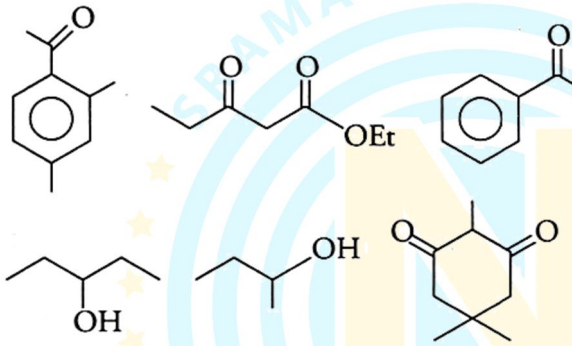
C) Fe^{3+}, Cr^{2+} D) Ti^{3+}, Mn^{2+}

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

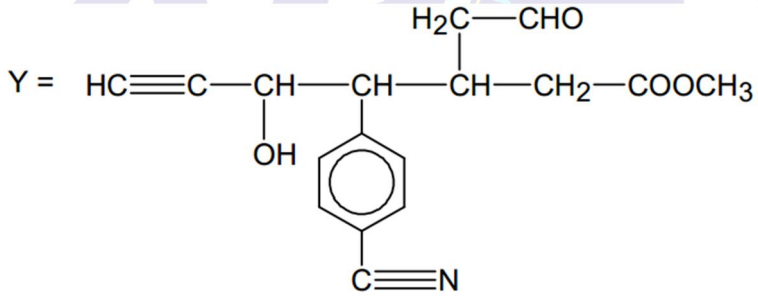
46. The number of molecules which gives haloform test among the following molecules is _____



47. A solution is prepared by adding 1 mole ethyl alcohol in 9 mole water. The mass percent of solute in the solution is _____ (integer answer) (Given: Molar mass in g mol^{-1}; ethyl alcohol: 46, water : 18)

48. For a reaction A → P
Half-life measured for two different values of initial concentrations 5×10^{-3} M and 25×10^{-4} M are 1 and 8 hrs respectively. If initial concentration is adjusted to 1.25×10^{-3} M, what is the half-life? (in hrs)

49. Number of moles of CH_3MgBr which will react with one mole of compound 'Y':



50. 60 ml of *x* M $AgNO_3$ required for complete precipitation of Br^- ions present in 15 ml of 0.02 M solution of $\left[Co(NH_3)_5 Br\right]Br_2$ as silver bromide. The value of 100*x* is _____.

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If $a_r = \cos\frac{2r\pi}{9} + i\sin\frac{2r\pi}{9}; r = 1,2,3,....$ where $i = \sqrt{-1}$ then the determinant $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is equal to

A) $a_2a_6 - a_4a_8$ B) a_9 C) $a_1a_9 - a_3a_7$ D) a_5

52. The co-ordinates of the points A and B are respectively $(-3,2)$ and $(2,3)$. P and Q are points on the line joining A and B such that $AP = PQ = QB$. A square $PQRS$ is constructed on PQ as one side, the co-ordinates of R can be:

(A) $\left(-\frac{4}{3},\frac{7}{3}\right)$ (B) $\left(\frac{2}{3},\frac{13}{3}\right)$ (C) $\left(\frac{1}{3},\frac{8}{3}\right)$ (D) $\left(\frac{2}{3},1\right)$

53. The number of integral value of m less than 10 for which the equation $\frac{(1+m^2)x^2 - 2(1+3m)x + (1+8m)}{2x^2+6x+9} > 0 \forall x \in R$:

(A) 9 (B) 3 (C) 0 (D) ∞

54. Let $f(x) = Max\left\{x^2, (1-x)^2, 2x(1-x)\right\}$ where $0 \le x \le 1$. Then the area of the region bounded by the curve $y = f(x)$, the x-axis and the lines $x = 0, x = 1$ is

(A) $\frac{17}{1}$ (B) $\frac{17}{27}$ (C) $\frac{14}{17}$ (D) $\frac{14}{27}$

55. The length of the latus-rectum of the ellipse, whose foci are $(2,5)$ and $(2,-1)$ and eccentricity is $\frac{3}{5}$, is

A) $\frac{18}{5}$ B) $\frac{50}{3}$ C) $\frac{10}{3}$ D) $\frac{32}{5}$

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56. Let $f(x)=\dfrac{\tan^{-1}\left(1-\{x\}\right).\cos^{-1}\left(1-\{x\}\right).\sin^{-1}\left(1-\{x\}\right)}{\sqrt{2\{x\}}\left(1-\{x\}\right)^2}$, where $\{x\}$ denotes fractional part of x, then $\lim\limits_{x\to 0^+} f(x)$:

(A) $\dfrac{\pi}{2\sqrt{2}}$ (B) $\dfrac{\pi^2}{8}$ (C) $\dfrac{\pi}{8}$ (D) 1

57. It is given that $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = xA+B$

Where A and B are determinants of order 3 not involving x. Then det $(A+B)$ is equal to:

(A) 12 (B) 24 (C) 15 (D) 18

58. Line L_1 is parallel to vector $\vec{a}=-3\hat{i}+2\hat{j}+4\hat{k}$ and passes through a point $A(7,6,2)$ and Line L_2 is parallel to vector $\vec{b}=2\hat{i}+\hat{j}+3\hat{k}$ and passes through a point $B(5,3,4)$. Now a line L_3 parallel to vector $\vec{r}=2\hat{i}-2\hat{j}-\hat{k}$ intersect the line L_1 and L_2 at points C and D, respectively, then $\left|\overrightarrow{CD}\right|=$

(A) 12 (B) 8 (C) 7 (D) 9

59. The exhaustive set of range of the values of p for which the equation $\sin\cos^{-1}\left(\cos\left(\tan^{-1}x\right)\right)=p$ has a solution is

A) $\left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right]$ B) $[0,1)$ C) $\left[\frac{1}{\sqrt{2}},1\right)$ D) $(-1,1)$

60. Let $b_1, b_2, \ldots\ldots, b_n$ be a geometric sequence such that $b_1+b_2=1$ and $\sum\limits_{k=1}^{\infty} b_k=2$. Given that $b_2<0$, then the value of b_1 is

A) $2-\sqrt{2}$ B) $1+\sqrt{2}$ C) $2+\sqrt{2}$ D) $4+\sqrt{2}$

61. Consider a branch of the hyperbola $x^2-2y^2-2\sqrt{2}x-4\sqrt{2}y-6=0$ with vertex at the point A. Let B be one of the end points of its latus rectum and C be the focus of the hyperbola nearest to the point A, the area of the triangle ABC (in sq. unit) is:

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(A) $\sqrt{\frac{3}{2}}$ (B) $\sqrt{\frac{3}{2}}-1$ (C) $\sqrt{\frac{3}{2}}+1$ (D) $\sqrt{\frac{2}{3}}$

62. If the foot of the perpendicular from point (4, 3, 8) on the $L_1 : \frac{x-a}{l}=\frac{y-2}{3}=\frac{z-b}{4}, l\neq 0$ is (3, 5, 7), then the shortest distance between the line L_1 and line L_2 : $L_2 : \frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ is equal to

(A) $\sqrt{\frac{2}{3}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{6}}$

63. The value of $\int_{-2}^{2}\frac{\left(1+\sqrt{|x|-x}\right)3^x+\left(\sqrt{|x|-x}\right)3^{-x}}{3^x+3^{-x}}dx$ is equal to

A) $3-2\sqrt{2}$ B) $3+2\sqrt{2}$ C) 11 D) $\frac{14}{3}$

64. If the sum of the first n terms of the series $\sqrt{3}+\sqrt{75}+\sqrt{243}+\sqrt{507}+.....$ is $435\sqrt{3}$, then n equals

(A) 29 (B) 18 (C) 15 (D) 13

65. If $I_1=\int_0^1\left(1-x^{50}\right)^{100}dx$ and $I_2=\int_0^1\left(1-x^{50}\right)^{101}dx$; $I_2=\alpha I_1$, then α equal to:

(A) $\frac{5049}{5050}$ (B) $\frac{5051}{5050}$ (C) $\frac{5050}{5051}$ (D) $\frac{5050}{5049}$

66. A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively, is received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is $\frac{3}{4}$. If the signal received at station B is green, then the probability that the original signal was green is

(A) $\frac{3}{5}$ (B) $\frac{6}{7}$ (C) $\frac{20}{23}$ (D) $\frac{9}{20}$

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67. If k be an integer such that the triangle with vertices (k, –3k), (5, k) and (–k, 2) has area 28 sq. units. Then the orthocenter of this triangle is at the point

(A) $\left(1,\frac{3}{4}\right)$ (B) $\left(1,-\frac{3}{4}\right)$ (C) $\left(2,\frac{1}{2}\right)$ (D) $\left(2,-\frac{1}{2}\right)$

68. If P, q ∈ {1, 2, 3, 4, 5}, then number of equations of the form $P^2x^2+q^2x+1=0$ having real roots is

(A) 20 (B) 16 (C) 14 (D) 12

69. Consider two vectors $\vec{u}=3\hat{i}-\hat{j}$ and $\vec{v}=2\hat{i}+\hat{j}-\lambda\hat{k},\lambda>0$. The angle between them is given by $\cos^{-1}\left(\frac{\sqrt{5}}{2\sqrt{7}}\right)$. Let $\vec{v}=\vec{v}_1+\vec{v}_2$, where $\vec{v}_1$ is parallel to $\vec{u}$ and $\vec{v}_2$ is perpendicular to $\vec{u}$. Then the value $3\left|\vec{v}_1-\vec{v}_2\right|^2+6\vec{v}_1.\vec{v}_2$ is equal to

A) 40 B) 42 C) 41 D) 10

70. Let $f(x)=\frac{e^{\tan x}-e^x+\ln(\sec x+\tan x)-x}{\tan x-x}$ be a continuous function at x = 0. The value of f(0) equals

A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{3}{2}$ D) 2

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SECTION – II
(Numerical Value Answer Type)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. The number of function f from the set $A=\{0,1,2\}$ into the set $B=\{0,1,2,3,4,5,6,7\}$ such that $f(i)\le f(j)$ for $i<j$ and $i, j\in A$ are equal to_____

72. Let $A=\{6,11,16,.....\}$ and $B=\{16,23,30,.....\}$ be the set consisting of the first 2024 terms of two arithmetic progressions. Then $n(A\cup B)$ is

73. Coefficient of x^6 in $\left((1+x)(1+x^2)^2(1+x^3)^3....(1+x^n)^n\right)$ where $n\ge 6$ is

74. Let $f(x)$ be a polynomial of degree 3 such that $f(-1)=10, f(1)=-6, f(x)$ has a critical point at $x=-1$ and $f'(x)$ has a critical point at x = 1. Then $f(x)$ has a local minima at x = _____

75. If $f(1)=1, f'(1)=3$, then the derivative of $f(f(f(x)))+(f(x))^2$ at $x=1$ is

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